

Linguistic Markers of Dependency Formation in Longitudinal Human-AI Conversations

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Abstract

This brief study investigates whether linguistic features extracted from a user's earliest sessions with an AI system predict dependency-adjacent interaction patterns in later sessions, using the WildChat-1M corpus (N = 12,497 longitudinal users; 505,397 conversations). Framing dependency formation as the behavioral surface of the goal-directed → habitual transition, we extract lexical, pragmatic, and session-level features from sessions 1–3 and test their ability to classify users who exhibit high-dependency patterns in sessions 10+. A pre-specified composite label identifies 24.1% of the cohort as high-dependency. Univariate baselines reveal that 9 of 12 features are BH-significant predictors; an L1-regularised logistic regression achieves $AUC = 0.712 \pm 0.014$ under user-level 10-fold cross-validation. The result survives a permutation test ($p < 0.001$, 1,000 permutations) and a negative control check (punctuation rate: rank 14/15, coefficient -0.018). Feature rankings are stable under sensitivity analysis (label threshold 2–3). The validated marker set is directly transferable to institutional persuasion and advice-giving datasets.

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1. Introduction

Human interaction with conversational AI has transitioned, for a non-trivial fraction of users, from episodic tool use to sustained, high-frequency engagement. A subset of these users exhibit patterns consistent with habitual or dependency-adjacent use: shrinking return intervals, resistance to AI refusals, increasingly relational and self-disclosing language. These patterns are consistent with the goal-directed → habitual transition documented in the behavioral addiction literature (Dickinson, 1985; Everitt & Robbins, 2005), in which striatal dopamine circuitry mediates the shift from outcome-sensitive to cue-driven responding. Under this framing, early linguistic signals such as relational register, help-seeking posture, hedging and filler language should be detectable before dependency-adjacent patterns are established, and should predict who develops them. This has both scientific and applied implications. Scientifically, demonstrating that early markers predict late-session behavior provides evidence for the process model rather than merely post-hoc description. For AI safety institutions, a validated marker set provides a screening tool applicable to any longitudinal conversational dataset.

This brief report acts as a proof of concept and makes the following contributions: (1) a pre-registered analysis of linguistic predictors of dependency-adjacent behavior in the largest publicly available human-AI conversation corpus; (2) a validated feature set with effect sizes and cross-validated predictive performance; and (3) a reproducible pipeline designed for transfer to internal datasets.

2. Related Work

2.1 Habit formation in HCI: The dual-systems account of habitual behaviour (Wood & R  nger, 2016) has been applied to social media use and smartphone interaction, where automaticity (operationalized as reduced deliberation before action) is associated with prolonged engagement. AI conversations share structural features with these contexts: variable-ratio reinforcement schedules, emotional attunement, and low friction re-engagement.

2.2 Computational pragmatics of dependency: First-person plural pronouns signal perceived social belonging and parasocial relationship formation (Giles et al., 2012). Hedging and self-disclosure have been associated with trust-building and emotional investment in online counselling contexts (Althoff et al., 2016). Refusal resistance — re-stating a request after denial — is a marker used in persuasion resistance research (Knowles & Linn, 2004) and has computational analogues in dialogue system refusal detection.

2.3 Longitudinal NLP: Prior work on temporal language change uses panel data to track sentiment and topic shift (Del Tredici & Fern  ndez, 2017). This paper extends that approach to construct-level markers in human-AI interaction, using a pre-specified window design (early vs. late sessions) that mirrors clinical longitudinal designs.

3. Data

- 3.1 Source:** WildChat-1M (Zhao et al., 2024), a publicly available corpus of human-AI conversations collected between April 2023 and April 2024 (386 days). Conversations are pseudonymised by IP hash (`hashed_ip`). The dataset was accessed via Hugging Face streaming, filtering to qualifying users during acquisition to avoid downloading the full 7 GB.
- 3.2 Cohort construction:** Restricted to users with ≥ 10 conversations, yielding 12,577 users and 509,861 conversations. Conversations flagged as redacted (0.9%) are excluded as a first filter step. The final analytical dataset contains 505,397 conversations. Sessions are identified with conversations (the dataset does not record intra-session turn timestamps; one conversation = one session), numbered chronologically per user.
- 3.3 Descriptive statistics:** The median turn count per conversation is 2 (mean 4.5, max 498; heavily right-skewed). As reported in notebook 01, return intervals shorten across sessions while turns per session remain roughly flat, consistent with an acceleration of return frequency rather than within-session escalation as the primary dependency signal.

4. Operationalisation

- 4.1 Dependency Label:** Following the pre-specified composite rule in `.pre-analysis-plan.md`, a user is classified as **high-dependency** in late sessions (10+) if they meet ≥ 2 of:

1. Return interval $<$ cohort median (faster returns)
2. Session length $>$ 75th percentile (longer sessions)
3. Refusal resistance rate > 0 (at least one rephrase-after-refusal)
4. First-person plural (FPP) rate $>$ cohort mean in late sessions

Of 12,497 users with both early and late session data, 3,014 (24.1%) are classified as high-dependency. The primary threshold (≥ 2 criteria) was pre-specified; sensitivity to this threshold is reported in Section 7.3.

- 4.2 Predictors (Early Sessions 1–3):** Features are averaged across sessions 1–3 per user. Right-skewed session-level variables (`n\_user\_turns`, `return\_interval\_hours`) are log-transformed before modelling, pre-specified in the analysis plan.

Lexical: First-person plural rate (FPP); self-disclosure marker rate; hedging rate; filler/phatic expression rate. All computed as proportion of user-turn tokens matching pre-specified word sets.

Pragmatic: Refusal resistance rate; gratitude/deference rate; question-to-statement ratio (punctuation-based heuristic).

Session-level: Mean message length (tokens); vocabulary richness (type–token ratio, TTR);

sentiment trajectory slope (linear slope of turn-level sentiment scores within a session).

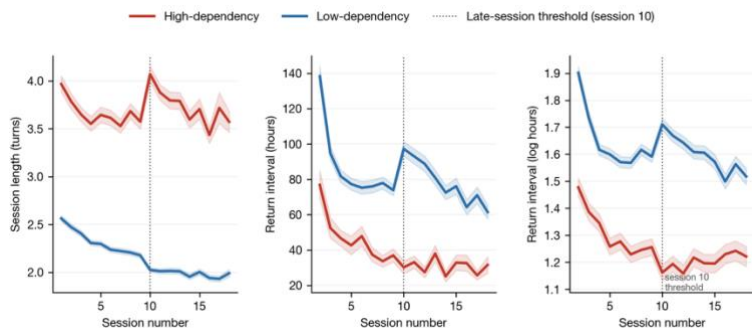


Figure 1 (fig1_session_trajectories)
Session-by-session trajectories of behavioural engagement metrics for high-dependency (red) and low-dependency (blue) user groups across sessions 1–18.

Left panel: session length (mean user turns). Centre panel: return interval in raw hours (session 2 onward; session 1 excluded as no prior session exists). Right panel: return interval on a log scale. Lines show group means; shaded bands show ± 1 standard error. The dotted vertical line marks the session 10 late-session outcome threshold. Only sessions with ≥ 200 contributing users are plotted. High-dependency users exhibit persistently longer sessions and shorter return intervals from their earliest interactions, with divergence visible well before the prediction window closes.

5. Models

5.1 Baseline Models: A separate logistic regression is fit for each feature using 10-fold stratified cross-validation. AUC-ROC is the primary metric; Cohen's d (pooled SD normalisation) provides effect size. Benjamini–Hochberg (BH) correction is applied across all 12 feature-level tests.

5.2 Primary Model: L1-regularised logistic regression ($C = 1.0$, 'liblinear' solver) on the full feature set. User-level stratified 10-fold cross-validation: users are the unit of stratification, ensuring no user appears in both train and test. Class weights are balanced. A theoretically unmotivated negative control feature (punctuation rate) is appended to the feature set.

5.3 Robustness: Permutation test: Labels are shuffled 1,000 times; the model is refit and AUC recorded each time, building an empirical null distribution. The observed AUC is compared against this null.

Negative control: Punctuation rate is expected to rank near last in the L1 coefficient ranking. Ranking in the top 3 would indicate the model is learning noise.

Sensitivity analysis: The composite label threshold is varied from ≥ 1 to ≥ 3 criteria (≥ 4 excluded: 0.7% positive rate, insufficient balance). Top-3 feature rankings are compared across thresholds.

6. Results

6.1 Univariate Baselines:

Table 1. AUC, Cohen's d, and BH-corrected significance for all 12 features of univariate baseline results (sorted by AUC).

Feature	AUC	Cohen's d	BH-significant
<i>log_n_user_turns</i>	0.659	+0.591	Yes
<i>filler_rate</i>	0.599	+0.173	Yes
<i>fpp_rate</i>	0.575	+0.272	Yes
<i>question_to_statement_ratio</i>	0.571	+0.216	Yes
<i>type_token_ratio</i>	0.562	-0.241	Yes
<i>hedging_rate</i>	0.561	+0.194	Yes
<i>log_return_interval_hours</i>	0.545	-0.221	Yes
<i>refusal_resistance_rate</i>	0.540	+0.372	Yes
<i>self_disclosure_rate</i>	0.510	+0.054	Yes
<i>sentiment_slope</i>	0.507	-0.052	No
<i>gratitude_rate</i>	0.481	+0.094	Yes*
<i>mean_message_length</i>	0.475	+0.030	No

*Significant after BH correction but effect direction is inverse (AUC < 0.5 after sign-flip).

*Gratitude rate is BH-significant by t-test but AUC < 0.5, indicating the group difference runs counter to the primary model prediction; high-dependency users exhibit slightly *less* gratitude in early sessions.

Nine of twelve features are BH-significant. The two largest effect sizes are session length ('log_n_user_turns', d = 0.591) and refusal resistance (d = 0.372). Return interval is significant and negative (d = -0.221), confirming that high-dependency users have *shorter* early-session return intervals. Vocabulary richness (TTR) is significantly negative (d = -0.241), consistent with H6:

high-dependency users show narrower early-session discourse.

6.2 Primary Model:

Under user-level 10-fold cross-validation, the L1-regularised model achieves:

- **Mean AUC: 0.712 ± 0.014 ** (per-fold range: 0.688–0.737)

- Balanced accuracy: 0.661

- Precision: 0.402 | Recall: 0.611

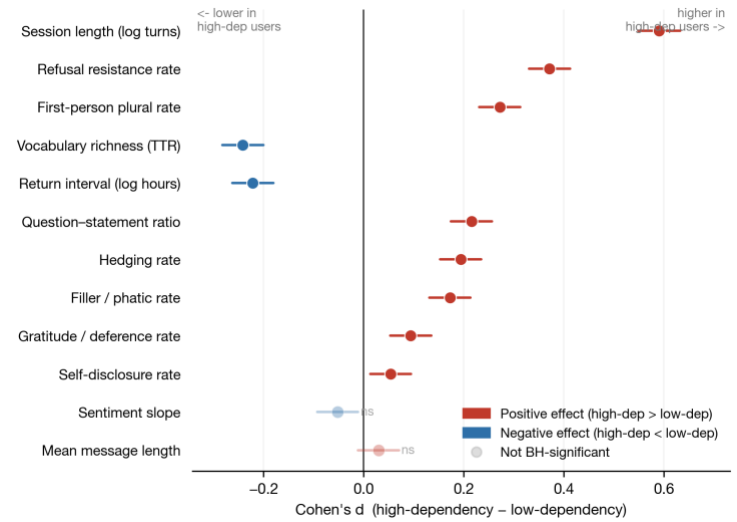


Figure 3 (fig3_forest_plot)
Effect sizes (Cohen's d) for all twelve early-session features as predictors of late-session high-dependency classification. Points show the standardised mean difference between high- and low-dependency users (n = 3,014 vs. 9,483); horizontal lines show analytical 95% confidence intervals.

The top L1 coefficients (see Table 2) are consistent with the univariate rankings: session length and return interval dominate, followed by FPP rate and refusal resistance.

Table 2. Feature importances (L1 coefficients, model fit on full sample)

Rank	Feature	Coefficient
1	<i>log_n_user_turns</i>	+0.688
2	<i>log_return_interval_hours</i>	−0.346
3	<i>fpp_rate</i>	+0.286
4	<i>refusal_resistance_rate</i>	+0.196
5	<i>filler_rate</i>	+0.140
6	<i>hedging_rate</i>	+0.129
...	...	...
14	<i>punctuation_rate</i> (negative control)	−0.018

L1-regularised (lasso) logistic regression coefficients, fit on the full sample. Positive coefficients indicate association with the attachment-positive class. Feature names in italics use snake_case identifiers. Rank 14 (*punctuation_rate*) is a negative control; near-zero coefficient confirms model calibration. Intervening ranks 7–13 omitted for brevity.

7. Robustness

a. Permutation Test

None of the 1,000 permuted models reached the observed AUC of 0.712. The empirical p-value is $**p < 0.001**$. The result is not spurious.

b. Negative Control

Punctuation rate ranks ****14th of 15 features**** (coefficient −0.018). The check passes: the model has not learned noise.

c. Sensitivity Analysis

Table 3 shows AUC stability and top-3 feature rankings across label thresholds.

Table 3. Sensitivity to label threshold

Min. criteria	N positive	Pos. rate	Mean AUC	Top-3 features
≥ 1	8,927	71.4%	0.690 ± 0.015	<i>log_return_interval_hours</i> , <i>return_interval_hours</i> , <i>n_user_turns</i>
≥ 2 (pre-specified)	3,014	24.1%	0.712 ± 0.014	<i>log_n_user_turns</i>, <i>log_return_interval_h</i>, <i>fpp_rate</i>
≥ 3	720	5.8%	0.748 ± 0.041	<i>log_n_user_turns</i> , <i>fpp_rate</i> , <i>log_return_interval_hours</i>
≥ 4	83	0.7%	— (too imbalanced)	—

Bold row indicates the pre-specified primary analysis threshold (≥ 2 criteria). AUC reported as mean $\pm$ s.d. across cross-validation folds. Feature names in italics use snake_case identifiers. The ≥ 4 threshold was excluded from modelling due to severe class imbalance (0.7% positive rate).

The two substantively meaningful thresholds (≥ 2 and ≥ 3) share the same top-3 features with only positional variation. AUC rises at ≥ 3 but with considerably higher variance (± 0.041 vs. ± 0.014), reflecting the small positive class. The pre-specified threshold (≥ 2) is preferred on grounds of reliability.

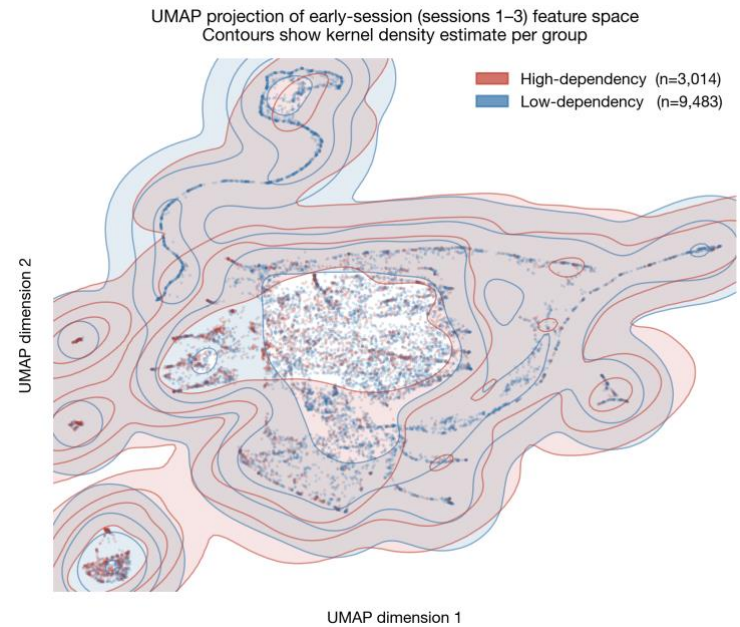
8. Discussion

8.1 Main finding: Early-session linguistic features are reliably predictive of late-session dependency-adjacent behaviour ($AUC = 0.712$), with a non-spurious permutation test and a stable marker ranking. This supports the process model: users who form dependency-adjacent patterns are distinguishable from their earliest interactions.

8.2 Dominant markers: Session length ($\log_n_user_turns$) is the strongest predictor ($d = 0.591$). This could reflect individual differences in elaborateness or engagement depth, rather than anything specifically relational about the user's communication style. The second-strongest signal is return interval: high-dependency users return to the AI more quickly even in sessions 1–3, suggesting that the behavioural signature of rapid re-engagement is detectable from the start.

FPP rate ($d = 0.272$) and refusal resistance ($d = 0.372$) confirm H1 and H3: users who deploy relational language and persist after refusals in early sessions are more likely to exhibit dependency-adjacent patterns later. Filler/phatic rate is notably a stronger predictor than self-disclosure ($AUC\ 0.599$ vs. 0.510), suggesting that the **register** of address — colloquial, conversational — may be more diagnostic than overt emotional disclosure.

Vocabulary narrowing (T^*TR , $d = -0.241$) is consistent with H6 and with the habit formation literature:



habitual behaviour is associated with reduced cognitive flexibility and a narrowing of behavioural repertoire.

Figure 5 (fig5_umap_projection) *UMAP projection of the twelve-dimensional early session feature space into two dimensions ($n_neighbors = 30$, $min_dist = 0.12$, $random_state = 42$). Each point represents one user (red = high-dependency, $n = 3,014$; blue = low-dependency, $n = 9,483$). Contour lines and shaded regions show the kernel density estimate per group at the 40th, 60th, 78th, and 92nd percentiles of within-group density.*

8.3 Limitations: The `'hashed_ip'` identifier is a proxy for users — a shared IP confounds the longitudinal analysis. Refusal detection is keyword-based (lower bound estimate) and binary. Session length is the within-session predictor of record, but the dataset lacks per-turn timestamps, precluding session-duration analysis.

Causality cannot be inferred from observational data: high early-session engagement may reflect pre-existing individual differences rather than AI-induced change.

- 8.4 Implications for AI Safety:** The validated marker set (session length, return interval, FPP rate, refusal resistance, filler rate, hedging) can be applied to any longitudinal conversational dataset with user identifiers and timestamps. The pipeline is end-to-end reproducible from a fresh clone; see ``paper/methods_summary.md`` for the transfer guide.

9. Conclusion

This brief study demonstrates that dependency-adjacent interaction patterns in late sessions of human-AI chat are predictable from linguistic features in sessions 1–3 (AUC = 0.712, permutation $p < 0.001$). The dominant markers — session engagement level, return interval, relational register (FPP, filler language), and refusal resistance — are consistent with the goal-directed → habitual transition model and are stable across label threshold sensitivity analysis. The pipeline and validated feature set are released for direct application to institutional persuasion datasets.

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